Task 3: Attrition Baseline Models and Evaluation

Documentation | Prepared by Katarukonda Ashok (GM00154) | October 2026

1. What this task is, in plain words

A **baseline** is the first honest score. Every smarter model built later has to beat it, otherwise we cannot prove an improvement. We predict who will leave within 12 months using simple, explainable models: a dummy that always says "stays", a logistic regression and a shallow decision tree, each with and without class weights. All of them use the Task 2 feature pipeline.

The challenge: only 14.6% of employees leave. A model that says "nobody leaves" is **85.4% accurate** and finds no one, so accuracy is the wrong yardstick. We use precision, recall, F1 and ROC-AUC.

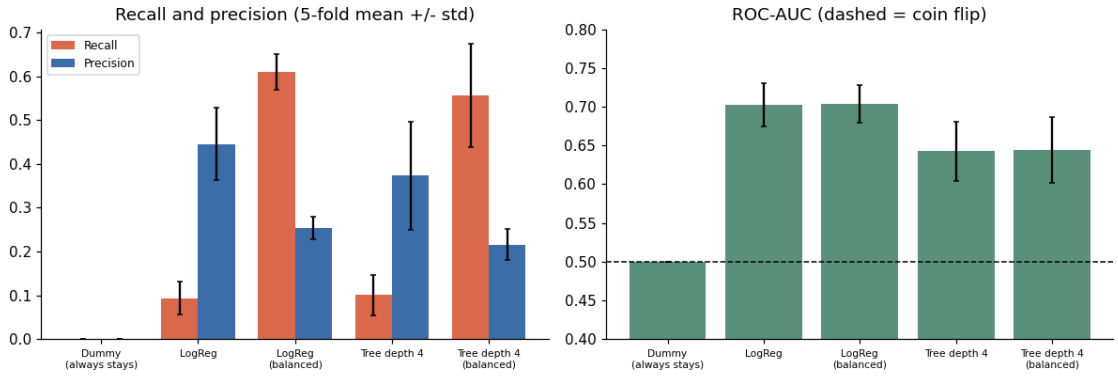
2. How it was done (and the rules that keep it honest)

- Stratified 80/20 split (1,906 train, 477 test). The **test set is locked** and used exactly once, at the end.
- Each model is *feature pipeline + model* in one object, so cross-validation refits the cleaning, caps, medians and encodings **inside every fold**.
- 5-fold stratified cross-validation; every metric is reported as mean +/- spread (standard deviation over the 5 folds).
- Class weights (balanced) tried for both model types and compared with the plain versions.

3. Results: cross-validated baseline table

Model	Precision	Recall	F1	ROC-AUC	Accuracy
Dummy (always stays)	0.000 +/- 0.000	0.000 +/- 0.000	0.000 +/- 0.000	0.500 +/- 0.000	0.854 +/- 0.001
Logistic Regression	0.446 +/- 0.083	0.093 +/- 0.038	0.152 +/- 0.051	0.702 +/- 0.028	0.850 +/- 0.004
Logistic Regression (balanced)	0.254 +/- 0.026	0.609 +/- 0.041	0.358 +/- 0.031	0.703 +/- 0.024	0.679 +/- 0.032
Decision Tree depth 4	0.374 +/- 0.124	0.101 +/- 0.047	0.156 +/- 0.069	0.643 +/- 0.038	0.844 +/- 0.012
Decision Tree depth 4 (balanced)	0.215 +/- 0.035	0.556 +/- 0.118	0.306 +/- 0.041	0.644 +/- 0.042	0.624 +/- 0.105

How to read it. *Precision*: of the people flagged as leavers, how many truly left. *Recall*: of the people who truly left, how many we flagged. *ROC-AUC*: how well the model ranks leavers above stayers (0.5 is a coin flip). The plain models look accurate only because they rarely predict "leaves" (recall near 0.1).

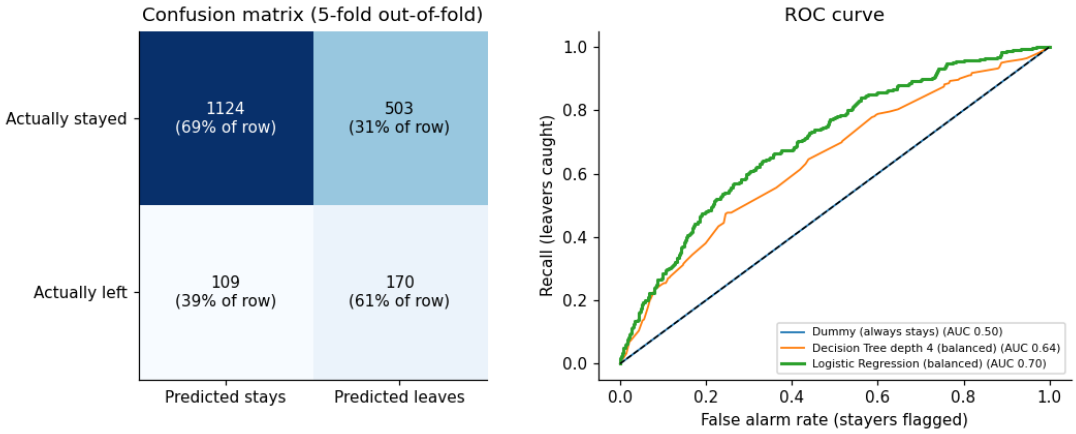


4. Class weights: what changed

Setting `class_weight="balanced"` tells the model that missing a leaver costs about 6 times more than a false alarm. For logistic regression, recall rises from 0.09 to 0.61, precision falls from 0.45 to 0.25, and ROC-AUC stays near 0.70. Weights move the cut-off, not the ranking: the same people are ranked highest, but more of them are flagged.

Chosen baseline: Logistic Regression (balanced). Rule written before touching the test set: among balanced models within 0.01 ROC-AUC of the best, take the one with the highest recall. It also beats the decision tree on ROC-AUC (0.70 vs 0.64) and is easier to explain.

5. Confusion matrix and ROC curve



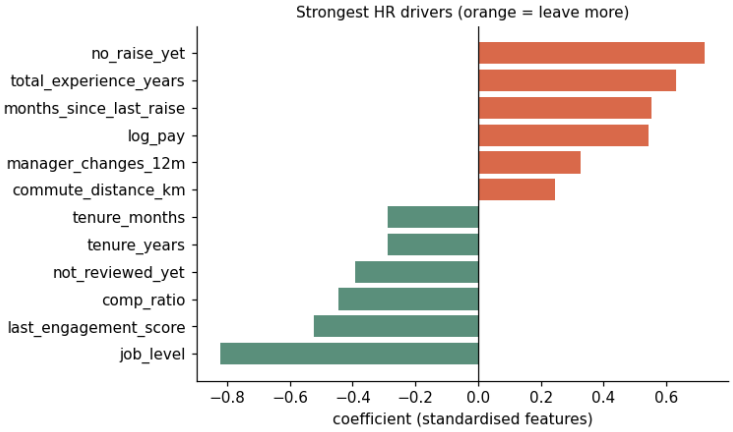
Out-of-fold on the training rows: the model caught **170 of 279** leavers (61%) and wrongly flagged 503 of 1627 stayers (31%). The bottom-left cell (missed leavers) is the costly mistake; the top-right cell is a wasted conversation.

6. What it means for HR

HR cannot talk to everyone. If they only review the highest-risk people, the model finds far more leavers than random picking would:

Review the top...	People reviewed	Leavers found	Share of all leavers found	Hit rate in that group	Better than random by
5% highest risk	95	38	14%	40%	2.7x
10% highest risk	191	63	23%	33%	2.3x
20% highest risk	381	115	41%	30%	2.1x
30% highest risk	572	153	55%	27%	1.8x

The model is a **screening aid**, not a verdict: the typical hit rate in the highest-risk group is about 30-40%, so most flagged people will not leave. Use it to prioritise conversations, never to judge an individual.



Readable drivers (logistic regression): people who have **never had a raise** or have waited long since the last one, who have changed managers more often, who have longer total careers, or who commute farther are more likely to leave. Higher job level, higher engagement score, a better compensation ratio (pay versus role midpoint) and longer tenure point toward staying. Pay-related columns overlap, so do not over-read any single coefficient. Department matters too (highest: D09, D06; lowest: D10, D05). These are associations in this dataset, not proven causes.

7. Think for yourself: precision or recall?

Recall matters most. A leaver we miss is a resignation with no chance to act. A stayer we flag by mistake costs one friendly conversation. But precision still has a job: if most flags are wrong, HR stops trusting the list, so we watch both and show the trade-off:

Flag if risk score at least	Share of staff flagged	Recall	Precision
0.3	61%	85%	20%
0.4	47%	71%	22%
0.5	35%	61%	25%
0.6	24%	47%	29%
0.7	14%	30%	31%

8. Final check on the locked test set (used once)

Metric	Cross-validation (mean +/- std)	Locked test set
Precision	0.254 +/- 0.026	0.227
Recall	0.609 +/- 0.041	0.586
F1	0.358 +/- 0.031	0.327
ROC-AUC	0.703 +/- 0.024	0.691

Test confusion matrix [[TN, FP], [FN, TP]] = [[267, 140], [29, 41]]. The test scores sit within the cross-validation spread (yes), so the baseline numbers are trustworthy and nothing leaked.

Time-based robustness: training on 2021-2024 observations and testing on 2025 gives ROC-AUC 0.71, recall 0.41 and precision 0.26 (1,898 train rows, 485 test rows). Ranking quality holds on newer data; recall at the 0.5 cut-off is lower, so the cut-off should be reviewed periodically.

9. How to run it

```
jupyter nbconvert --to notebook --execute --inplace Task3_Baseline_Models/task3_baseline_models.ipynb
```

Code: `src/models/baselines.py` (model list, cross-validation, lift table). Outputs: `Task3_Baseline_Models/outputs/` (tables and figures). Reference scores in the workbook's baseline_benchmarks sheet (logistic regression ROC-AUC about 0.71, recall about 0.63) are consistent with these results.

10. Completion criteria

Completion criterion	Status	Evidence
Dummy, logistic and tree baselines compared	Done	Table in section 3 (plain and balanced versions)
Cross-validated metrics saved	Done	outputs/tables/cv_summary_full.csv, cv_per_fold.csv, baseline_table.csv
Test set untouched until final check	Done	Used once in Step 9 of the notebook
Report written for non-technical readers	Done	This document: sections 1, 6 and 7

Next: Task 4 changes the kind of problem: finding unusual rows without labels.